

# NEXUS-R Naming Consistency Guide

Clarity over cleverness. Predictability over novelty.

## Core Principles

1. **Verb-object for actions** — `parse_intent`, `route_task`, `verify_step`
2. **Noun-descriptor for data** — `intent_result`, `routing_decision`, `audit_entry`
3. **Subsystem prefix for module files** — `ig_`, `car_`, `es_`, `etd_`
4. **No abbreviations without glossary entry** — ETD is defined once, used everywhere
5. **American English** — `behavior`, not `behaviour`; `color`, not `colour`

## Approved Glossary (Use These Terms)

| Term                      | Meaning                                                         | Never Use Instead                                   |
|---------------------------|-----------------------------------------------------------------|-----------------------------------------------------|
| <b>ETD</b>                | Execution Trace<br>Distillation                                 | “cognitive compression”, “workflow cache”, “memory” |
| <b>CAR</b>                | Capability-Aware Router                                         | “model router”, “LLM selector”, “tier picker”       |
| <b>CPC</b>                | Cache of Proven<br>Capabilities (legacy) /<br>CPC Short-Circuit | “cache check”, “fast path”                          |
| <b>ISE</b>                | Intent-Specific Execution<br>(legacy, now part of CAR)          | —                                                   |
| <b>MCP</b>                | Model Context Protocol                                          | “tool protocol”, “custom connector”                 |
| <b>T1-T4</b>              | Task tiers (read-only →<br>irreversible)                        | “levels”, “classes”, “priorities”                   |
| <b>BYOK</b>               | Bring Your Own Key<br>(cloud provider)                          | “cloud”, “premium”, “paid”                          |
| <b>EventStore</b>         | SQLite-backed append-only<br>event log                          | “database”, “log”, “history”                        |
| <b>WorkingState</b>       | In-memory volatile task<br>context                              | “context”, “session”, “state”                       |
| <b>IdentityStore</b>      | Encrypted user preferences<br>and keys                          | “profile”, “config”, “settings”                     |
| <b>CausalEvent</b>        | Event with<br><code>parent_event_id</code> linkage              | “linked event”, “chain event”                       |
| <b>IntentResult</b>       | Parsed user intent with<br>metadata                             | “parse result”, “user request”                      |
| <b>RoutingDecision</b>    | Selected model, tier, cost<br>estimate                          | “model choice”, “selection”                         |
| <b>VerificationResult</b> | Step pass/fail with detail                                      | “check result”, “validation”                        |
| <b>PermissionDecision</b> | Approve/deny with tier<br>and reason                            | “auth result”, “access”                             |
| <b>SecretRegistry</b>     | OS credential manager<br>wrapper                                | “key store”, “password manager”                     |

Table 1 – continued

| Term                 | Meaning                        | Never Use Instead              |
|----------------------|--------------------------------|--------------------------------|
| <b>Sandbox</b>       | Isolated execution environment | “container”, “jail”, “env”     |
| <b>TelemetrySpan</b> | Structured timing/metric event | “log entry”, “metric”, “trace” |

## File Naming Conventions

### Foundation Layer

```
# Correct
foundation/nexus_r/events.py
foundation/nexus_r/models.py
foundation/nexus_r/config.py
foundation/nexus_r/exceptions.py
foundation/nexus_r/telemetry.py

# Incorrect
foundation/core/event_system.py
foundation/utils/model_stuff.py
foundation/settings.py
foundation/errors.py
foundation/logging.py
```

### Module Layer

```
# Correct
modules/input_gateway/src/intent_parser.py
modules/cognition_router/src/router.py
modules/execution_sandbox/src/sandbox.py
modules/state_core/src/event_store.py
modules/workflow_engine/src/trace_recorder.py
modules/trust_layer/src/permission_enforcer.py
modules/session_manager/src/manager.py

# Incorrect
modules/input/src/parser.py # missing subsystem clarity
modules/router/src/main.py # too generic
modules/sandbox/src/exec.py # abbreviation
modules/state/src/store.py # too generic
modules/etd/src/recorder.py # use full subsystem name
modules/security/src/auth.py # trust_layer is the spec name
```

### Test Layer

```
# Correct
tests/unit/test_input_gateway.py
```

```

tests/unit/test_intent_parser.py
tests/unit/test_cognition_router_adversarial.py
tests/integration/test_full_pipeline.py
tests/security/test_sandbox_hardening.py
tests/stress/test_concurrency_100.py

# Incorrect
tests/test_ig.py # abbreviation
tests/integration/pipeline.py # missing test_ prefix
tests/security/injection.py # missing test_ prefix
tests/stress/big_load.py # non-descriptive

```

## Class Naming

| Pattern      | Example                                           | Used For                                           |
|--------------|---------------------------------------------------|----------------------------------------------------|
| Noun         | EventStore, ModelRouter, SessionManager           | Core services                                      |
| NounResult   | IntentResult, RoutingDecision, VerificationResult | Return structs                                     |
| NounError    | RoutingError, ExecutionError, SecurityError       | Exceptions                                         |
| NounConfig   | SandboxConfig, NEXUSConfig                        | Configuration                                      |
| AbstractNoun | —                                                 | Avoid. Use protocols or concrete names             |
| I Noun       | —                                                 | Avoid. No Hungarian notation                       |
| NounImpl     | —                                                 | Avoid. Names should describe role, not inheritance |

## Function Naming

| Pattern      | Example                               | Used For                  |
|--------------|---------------------------------------|---------------------------|
| verb_noun    | parse_intent, route_task, verify_step | Core actions              |
| get_noun     | get_profile, get_chain, get_cost      | Retrieval                 |
| set_noun     | set_preference, set_state             | Mutation                  |
| is_adjective | is_valid, is_expired, is_locked       | Boolean checks            |
| has_noun     | has_permission, has_session           | Possession checks         |
| check_noun   | check_permission, check_invalidation  | Validation                |
| compute_noun | compute_embedding, compute_cost       | Calculation               |
| handle_noun  | handle_error, handle_fallback         | Error paths               |
| do_verb      | —                                     | Avoid. Use specific verbs |

Table 3 – continued

| Pattern | Example | Used For           |
|---------|---------|--------------------|
| process | —       | Avoid. Too generic |
| manage  | —       | Avoid. Too generic |

## Variable Naming

| Context      | Pattern                   | Example                              |
|--------------|---------------------------|--------------------------------------|
| Loop indices | i, j, k                   | Acceptable only in math contexts     |
| Event IDs    | evt_ prefix               | evt_a1b2c3                           |
| Session IDs  | sess_ prefix              | sess_x9y8z7                          |
| ETD IDs      | etd_ prefix               | etd_deploy_vercel                    |
| Task IDs     | task_ prefix              | task_20260523_001                    |
| Timestamps   | ts_ prefix                | ts_created, ts_modified              |
| Costs        | cost_ prefix              | cost_estimate, cost_actual           |
| Counters     | count_ prefix             | count_success, count_failure         |
| Rates        | rate_ prefix              | rate_success,<br>rate_generalization |
| Durations    | duration_ms or latency_ms | duration_ms, latency_p99_ms          |

## Error Code Ranges

| Module            | Range            | Example                                |
|-------------------|------------------|----------------------------------------|
| Input Gateway     | IG-001 to IG-050 | IG-014: Invalid intent classification  |
| Cognition Router  | CR-001 to CR-050 | CR-031: Cost cap exceeded              |
| Execution Sandbox | ES-001 to ES-080 | ES-045: Sandbox escape detected        |
| State Core        | SC-001 to SC-050 | SC-022: Event chain broken             |
| Workflow Engine   | WE-001 to WE-100 | WE-067: Generalization below threshold |
| Trust Layer       | TL-001 to TL-100 | TL-089: T4 approval denied             |
| Session Manager   | SM-001 to SM-050 | SM-012: Session expired                |
| Foundation        | NX-001 to NX-050 | NX-001: Configuration invalid          |

## Directory Structure (Canonical)

```
nexus-r/
├─ specs/                # Source of truth. Read, never write.
└─ 00_architecture.md
```

```

├── 01_input_gateway.spec.md
├── 02_cognition_router.spec.md
├── 03_execution_sandbox.spec.md
├── 04_state_core.spec.md
├── 05_workflow_engine.spec.md
├── 06_trust_layer.spec.md
├── 03d_session_management.spec.md # Phase 3 extension
├── 99_acceptance_criteria.md
├── foundation/
│   ├── nexus_r/
│   │   ├── __init__.py
│   │   ├── events.py # Event, CausalEvent, EventStore
│   │   ├── models.py # ModelRouter, ProviderProfile
│   │   ├── config.py # NEXUSConfig
│   │   ├── exceptions.py # NEXUSError hierarchy
│   │   └── telemetry.py # TelemetrySpan, structured logging
├── modules/
│   ├── input_gateway/
│   │   ├── README.md
│   │   ├── interface.yaml
│   │   ├── src/
│   │   │   ├── __init__.py
│   │   │   ├── intent_parser.py
│   │   │   ├── classifier.py
│   │   │   └── parameter_extractor.py
│   │   └── tests/
│   │       ├── test_intent_parser.py
│   │       └── test_classifier.py
│   ├── cognition_router/
│   │   ├── README.md
│   │   ├── interface.yaml
│   │   ├── src/
│   │   │   ├── __init__.py
│   │   │   ├── router.py # CognitionRouter
│   │   │   ├── cpc_check.py # CPCShortCircuit
│   │   │   ├── capability_profiler.py
│   │   │   └── fallback.py # FallbackChain
│   │   └── tests/
│   │       └── test_router.py
│   ├── execution_sandbox/
│   │   ├── README.md
│   │   ├── interface.yaml
│   │   ├── src/
│   │   │   ├── __init__.py
│   │   │   ├── sandbox.py # ExecutionSandbox
│   │   │   ├── mcp_client.py # MCPClient
│   │   │   └── verifier.py # InlineVerifier

```

```

| | | └─ permission_checker.py
| | | └─ trace_logger.py      # TraceLogger
| | └─ tests/
| |   └─ test_sandbox.py
|
| └─ state_core/
| | └─ README.md
| | └─ interface.yaml
| | └─ src/
| | | └─ __init__.py
| | | └─ event_store.py      # Enhanced EventStore
| | | └─ working_state.py    # WorkingState
| | | └─ identity_store.py   # IdentityStore
| | | └─ projected_views.py  # ViewManager
| | └─ tests/
| |   └─ test_event_store.py
|
| └─ workflow_engine/
| | └─ README.md
| | └─ interface.yaml
| | └─ src/
| | | └─ __init__.py
| | | └─ trace_recorder.py   # TraceRecorder
| | | └─ trace_analyzer.py   # TraceAnalyzer (stub Phase 1)
| | | └─ distiller.py        # TraceDistiller (Phase 2)
| | | └─ generalizer.py      # GeneralizationVerifier (Phase 2)
| | | └─ parameterizer.py    # WorkflowParameterizer (Phase 2)
| | | └─ retriever.py        # ETDRetriever (Phase 2)
| | | └─ applicator.py       # ConservativeApplicator (Phase 2)
| | | └─ invalidator.py      # ETDInvalidator (Phase 2)
| | | └─ composer.py         # ETDComposer (Phase 3)
| | | └─ replayer.py         # ExecutionReplayer (Phase 3)
| | | └─ forker.py           # ExecutionForker (Phase 3)
| | └─ tests/
| |   └─ test_trace_recorder.py
|
| └─ trust_layer/
| | └─ README.md
| | └─ interface.yaml
| | └─ src/
| | | └─ __init__.py
| | | └─ permission_enforcer.py
| | | └─ audit_logger.py
| | | └─ cost_tracker.py
| | | └─ secret_handler.py   # SecretHandler
| | | └─ risk_classifier.py   # MLRiskClassifier (Phase 2)
| | | └─ prompt_injection_defense.py # PromptInjectionDetector (Phase 2)
| | └─ tests/
| |   └─ test_permission_enforcer.py
|
| └─ session_manager/      # Phase 3 extension

```

```

| | |─ README.md
| | |─ interface.yaml
| | |─ src/
| | |   |─ __init__.py
| | |   └─ manager.py          # SessionManager
| | └─ tests/
| |     └─ test_session_manager.py
| |
| └─ cli/
|     |─ README.md
|     |─ interface.yaml
|     └─ src/
|         |─ __init__.py
|         └─ main.py          # Typer CLI entry point
|
└─ tests/
    |─ unit/
    |─ integration/
    |─ security/
    |─ stress/
    |─ failure/
    └─ fixtures/
|
└─ scripts/
    |─ verify_module.py
    |─ phase1_benchmark.py
    |─ phase1_real_validation.py
    └─ phase1_failure_injection.py
|
└─ docs/
    |─ architecture_review_phase1.md
    |─ benchmark_report_phase1.md
    |─ security_audit_phase1.md
    |─ known_limitations_phase1.md
    |─ technical_debt_phase1.md
    |─ runtime_truthfulness_report_phase1_5.md
    |─ session-runtime-failure-report.md
    |─ persistence-redesign-proposal.md
    |─ recovery-architecture-proposal.md
    └─ concurrency-risk-report.md
|
└─ e2e/
└─ .github/
    └─ workflows/
        └─ ci.yml
└─ pyproject.toml
└─ README.md
└─ CONTRIBUTING.md
└─ ROADMAP.md
└─ ARCHITECTURE.md
└─ LICENSE

```

```
└─ .gitignore
```

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## Commit Message Style

| Type     | Format                       | Example                                                             |
|----------|------------------------------|---------------------------------------------------------------------|
| Feature  | feat(scope): description     | feat(car): add parallel probe for uncertain tasks                   |
| Fix      | fix(scope): description      | fix(sandbox): prevent path traversal via symlink                    |
| Test     | test(scope): description     | test(etd): add generalization verification at 90% threshold         |
| Docs     | docs(scope): description     | docs(spec): update CAR fallback chain to 6 tiers                    |
| Refactor | refactor(scope): description | refactor(events): replace per-write connection with connection pool |
| Security | security(scope): description | security(trust): add prompt injection anomaly detection             |
| Perf     | perf(scope): description     | perf(store): batch event appends, 100x improvement                  |

**Scope values:** foundation, gateway, router, sandbox, state, workflow, trust, session, cli, tests, docs, repo

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## Documentation Style

### For Specs

- Present tense, imperative mood
- Every function: signature, parameters, return type, error codes
- Every schema: JSON example with all fields
- No implementation details — only contracts

### For READMEs

- Second person (“You can...”, “Run...”)
- Code examples must be copy-pasteable
- Status badges must reflect reality

### For Reports

- Honest classification: REAL / PARTIAL / MOCK / UNVERIFIED
  - Numbers with methodology, not just results
  - Known limitations listed explicitly
-



Consistency is not bureaucracy. It is the respect you show to the next person who reads your code.